

18. Elements Dhundhna (getElement*, querySelector*)

Jab elements door door hon, to navigation ke bajay search methods use karte hain.

getElementById

```
js  
  
let elem = document.getElementById('elem');
```

- Sirf `document` par chalta hai (kisi element par nahi).
- `id` **unique** honi chahiye.
- Browser `id` naam ka global variable bhi bana deta hai (jaise `elem`), par isse **avoid** karo (naming conflicts ka risk). Local JS variable ho to wo is par priority leta hai.

querySelectorAll(css)

Sabse powerful. Kisi bhi **CSS selector** se **saare** matching elements deta hai.

```
js  
  
document.querySelectorAll('ul > li:last-child');
```

Pseudo-classes (`:hover`, `:active`) bhi chalte hain.

querySelector(css)

Sirf **pehla** matching element (`querySelectorAll(css)[0]`) jaisa, par tez.

elem.matches(css)

Kuch dhundhta nahi, sirf check karta hai ki `elem` selector se match karta hai ya nahi (`true/false`). Filter karne me kaam aata hai.

```
js  
  
if (elem.matches('a[href$=".zip"]')) { ... }
```

elem.closest(css)

Element se **upar** ki taraf (khud element bhi) sabse paas ka **ancestor** dhundhta hai jo selector se match kare. Na mile to `null`.

```
js
```

```
chapter.closest('.book'); // paas wala UL
```

getElementsBy* (purane tarike)

- `getElementsByTagName(tag)` (`'*' = sab`)
- `getElementsByClassName(className)`
- `document.getElementsByName(name)`

Ab zyaadatar history, par purani scripts me milte hain.

Galtiyan:

- `s` mat bhulo: `getElementsByTagName` (collection), `getElementById` (ek element).
- Ye **collection** dete hain, seedha `.value = 5` nahi kar sakte. Index lagao ya loop chalao:

```
js
document.getElementsByTagName('input')[0].value = 5;
```

Live vs Static

- `getElementsBy*` = **live** collection (DOM badle to update ho jaata hai).
- `querySelectorAll` = **static** (ek fixed snapshot).

Summary Table

Method	Kis se dhundhta hai	Element par call?	Live?
<code>querySelector</code>	CSS selector	Haan	Nahi
<code>querySelectorAll</code>	CSS selector	Haan	Nahi
<code>getElementById</code>	id	Nahi	Nahi
<code>getElementsByName</code>	name	Nahi	Haan
<code>getElementsByTagName</code>	tag	Haan	Haan
<code>getElementsByClassName</code>	class	Haan	Haan

Extra: `elemA.contains(elemB)` = `elemB`, `elemA` ke andar hai to `true` (ya dono same hon).

Yaad rakho

Zyaadatar kaam `querySelector` aur `querySelectorAll` se ho jaata hai. `matches` filter ke liye, `closest` upar ke ancestor ke liye.